

CORE CS (THEORY)

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❖ Top 50 Core CS Interview Questions

1. What is Computer Science?

Computer Science is the study of computation and information processing. It focuses on algorithms, data structures, software, and hardware interaction. It helps solve real-world problems using computers. It forms the foundation of IT and software development.

2. What is an Algorithm?

An algorithm is a step-by-step procedure to solve a problem. It must be finite and clear. Algorithms improve efficiency and accuracy. They are language-independent.

3. What is Data Structure?

A data structure organizes data efficiently. It allows fast access and modification. Examples include arrays, stacks, and trees. Proper data structures improve performance.

4. What is Time Complexity?

Time complexity measures algorithm execution time. It depends on input size. Expressed using Big-O notation. Helps compare algorithms.

5. What is Space Complexity?

Space complexity measures memory usage of an algorithm. Includes input and auxiliary space. Helps optimize memory usage. Important for large data.

6. What is Big-O Notation?

Big-O represents worst-case performance. It describes growth rate of algorithms. Examples include $O(n)$, $O(\log n)$. Used for analysis.

7. What is Operating System?

An OS manages hardware and software resources. It provides an interface to users. Handles memory, process, and files. Examples include Windows and Linux.

8. What is Process?

A process is a program in execution. It has its own memory and resources. OS schedules processes. Multiple processes can run concurrently.

9. What is Thread?

A thread is a lightweight process. Threads share memory within a process. They improve performance. Used in multitasking.

10. Difference between Process and Thread

A process has separate memory. Threads share memory. Process creation is costly. Threads are faster.

11. What is Deadlock?

Deadlock occurs when processes wait indefinitely. Resources are locked. No process can proceed. Common in multitasking systems.

12. What is Memory Management?

Memory management allocates and frees memory. OS handles it. Ensures efficient usage. Prevents memory leaks.

13. What is Virtual Memory?

Virtual memory uses disk as extra memory. Allows large programs to run. Improves utilization. Implemented using paging.

14. What is DBMS?

DBMS manages databases. It stores and retrieves data efficiently. Ensures security and consistency. Examples include MySQL and Oracle.

15. What is SQL?

SQL is used to query databases. It manages relational data. Supports CRUD operations. Standard database language.

16. What is Normalization?

Normalization reduces data redundancy. Improves data integrity. Data is divided into tables. Uses normal forms.

17. What is Primary Key?

Primary key uniquely identifies records. It cannot be NULL. Ensures uniqueness. Only one per table.

18. What is Foreign Key?

Foreign key links two tables. It refers to primary key. Maintains referential integrity. Prevents invalid data.

19. What is Computer Network?

A computer network connects multiple systems. Enables data sharing. Uses protocols. Examples include LAN and WAN.

20. What is Protocol?

Protocol defines communication rules. Ensures proper data transfer. Examples include TCP/IP. Used in networking.

21. What is TCP/IP?

TCP/IP is a networking protocol suite. TCP ensures reliable delivery. IP handles addressing. Used on the internet.

22. What is IP Address?

IP address uniquely identifies a device. Used for communication. Can be IPv4 or IPv6. Works at network layer.

23. What is DNS?

DNS converts domain names to IP addresses. Makes internet user-friendly. Works like phonebook. Essential for browsing.

24. What is Compiler?

Compiler converts source code to machine code. Entire program compiled at once. Faster execution. Examples include GCC.

25. What is Interpreter?

Interpreter executes code line by line. Easier debugging. Slower execution. Examples include Python interpreter.

26. Difference between Compiler and Interpreter

Compiler translates whole code. Interpreter translates line by line. Compiler is faster. Interpreter is flexible.

27. What is OOPS?

OOPS is programming paradigm. Based on objects and classes. Improves code reuse. Used in Java and C++.

28. What is Class?

Class is blueprint of objects. Defines properties and methods. No memory allocated. Used to create objects.

29. What is Object?

Object is instance of class. Has state and behavior. Memory is allocated. Used in OOPS.

30. What is Encapsulation?

Encapsulation wraps data and methods together. Protects data. Improves security. Uses access specifiers.

31. What is Inheritance?

Inheritance allows class to acquire properties. Promotes code reuse. Parent-child relationship. Improves extensibility.

32. What is Polymorphism?

Polymorphism means many forms. Same method behaves differently. Achieved via overloading and overriding. Improves flexibility.

33. What is Abstraction?

Abstraction hides implementation details. Shows essential features. Improves simplicity. Achieved using interfaces.

34. What is Stack?

Stack follows LIFO order. Supports push and pop. Used in recursion. Limited access structure.

35. What is Queue?

Queue follows FIFO order. Supports enqueue and dequeue. Used in scheduling. Efficient for processing.

36. What is Linked List?

Linked list stores nodes. Each node points to next. Dynamic size. Efficient insertion and deletion.

37. What is Tree?

Tree is hierarchical structure. Root node at top. Used in searching. Examples include BST.

38. What is Graph?

Graph consists of nodes and edges. Represents networks. Can be directed or undirected. Used in routing.

39. What is Recursion?

Recursion is function calling itself. Simplifies complex problems. Requires base condition. Uses stack memory.

40. What is Sorting?

Sorting arranges data in order. Improves searching efficiency. Examples include bubble and merge sort. Used widely.

41. What is Searching?

Searching finds an element. Can be linear or binary. Binary search is faster. Requires sorted data.

42. What is Hashing?

Hashing maps data to index. Improves access time. Uses hash functions. Used in hash tables.

43. What is Software Engineering?

Software engineering applies engineering principles. Develops quality software. Uses SDLC models. Ensures reliability.

44. What is SDLC?

SDLC is software development life cycle. Includes planning, design, testing. Ensures systematic development. Improves quality.

45. What is Testing?

Testing checks software defects. Ensures correctness. Improves reliability. Includes unit and integration testing.

46. What is Agile Model?

Agile uses iterative development. Focuses on customer feedback. Flexible approach. Used in modern projects.

47. What is Version Control?

Version control manages code changes. Supports collaboration. Tracks history. Examples include Git.

48. What is Cloud Computing?

Cloud computing provides online services. Includes storage and servers. Reduces infrastructure cost. Examples include AWS.

49. What is Cyber Security?

Cyber security protects systems from attacks. Ensures data safety. Uses encryption and firewalls. Important in digital world.

50. Why is Core CS Important?

Core CS builds strong fundamentals. Helps in problem-solving. Used in interviews. Essential for career growth.

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